



Getting Started with Azure: Master These 15 Services First



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1. Microsoft Entra ID (formerly Azure Active Directory)

What it is: Cloud-based identity and access management service.

Key Features:

- Single sign-on (SSO) across applications
- Multi-factor authentication (MFA)
- Conditional access policies
- Identity protection and governance
- B2B and B2C identity scenarios
- Role-based access control (RBAC)

When to use:

- Always - it's the foundation of Azure security
- User authentication and authorization
- Managing access to Azure resources
- SSO for SaaS applications
- Identity federation with on-premises AD

When NOT to use:

- You really can't avoid it in Azure

Official Documentation: <https://learn.microsoft.com/en-us/entra/identity/>

2. Azure OpenAI Service

What it is: Managed service providing access to OpenAI's models (GPT-4, GPT-3.5, DALL-E, Whisper, embeddings).

Key Features:

- Enterprise-grade security and compliance
- Private networking support
- Content filtering and responsible AI features
- Pay-per-token pricing
- Integration with Azure services
- Fine-tuning capabilities

When to use:

- Building AI-powered applications



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- Natural language processing tasks
- Code generation and analysis
- Content creation and summarization
- Chatbots and conversational AI
- Semantic search and embeddings

When NOT to use:

- Simple rules-based automation
- When cost per token is prohibitive
- Real-time low-latency requirements (consider edge solutions)
- Regions where service isn't available

Official Documentation: <https://learn.microsoft.com/en-us/azure/ai-services/openai/>

3. Azure AI Services (Cognitive Services)

What it is: Pre-built AI capabilities for vision, speech, language, and decision-making.

Key Features:

- Vision: Computer Vision, Face API, Custom Vision
- Speech: Speech-to-text, text-to-speech, translation
- Language: Text analytics, translation, Language Understanding (LUIS)
- Decision: Anomaly Detector, Content Moderator
- No ML expertise required
- REST APIs and SDKs

When to use:

- Adding AI capabilities without building models
- Image/video analysis and OCR
- Speech recognition and synthesis
- Sentiment analysis and language detection
- Content moderation

When NOT to use:

- Need highly customized AI models (use Azure Machine Learning)
- OpenAI models better suit the use case
- Real-time video processing at scale

Official Documentation: <https://learn.microsoft.com/en-us/azure/ai-services/>



4. Azure Virtual Machines (VMs)

What it is: Infrastructure-as-a-Service (IaaS) compute resources - cloud-hosted servers.

Key Features:

- Full control over OS and software stack
- Support for Windows and Linux
- Various VM sizes for different workloads
- Virtual machine scale sets for auto-scaling
- Integration with virtual networks and storage
- Spot VMs for cost savings

When to use:

- Lift-and-shift migrations from on-premises
- Need full OS control
- Running legacy applications
- Testing and development environments
- High-performance computing

When NOT to use:

- Simple web apps (use App Service instead)
- Want to avoid infrastructure management
- Cost-sensitive workloads that could use serverless

Official Documentation: <https://learn.microsoft.com/en-us/azure/virtual-machines/>

5. Azure App Service

What it is: Platform-as-a-Service (PaaS) for hosting web applications, APIs, and mobile backends.

Key Features:

- Auto-scaling and load balancing
- Built-in CI/CD integration
- Support for .NET, Java, Node.js, Python, PHP
- SSL certificates and custom domains
- Deployment slots for staging



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- WebJobs for background tasks

When to use:

- Web applications and REST APIs
- Need quick deployment without managing infrastructure
- Want built-in scaling and monitoring
- Mobile app backends

When NOT to use:

- Need OS-level control (use VMs)
- Non-HTTP workloads
- Extremely cost-sensitive sporadic traffic (consider Functions)

Official Documentation: <https://learn.microsoft.com/en-us/azure/app-service/>

6. Azure Storage Account

What it is: Scalable cloud storage service offering multiple storage types.

Key Features:

- Blob Storage: Object storage for unstructured data (hot, cool, archive tiers)
- File Storage: Managed file shares (SMB protocol)
- Queue Storage: Message queuing
- Table Storage: NoSQL key-value store
- Multiple redundancy options (LRS, GRS, ZRS, GZRS)
- Lifecycle management policies

When to use:

- Storing files, images, videos, backups
- Application data and logs
- Static website hosting
- Shared file systems
- Data lake storage (ADLS Gen2)

When NOT to use:

- Relational data (use SQL Database)
- Complex querying (use Cosmos DB)

Official Documentation: <https://learn.microsoft.com/en-us/azure/storage/>



7. Azure SQL Database

What it is: Fully managed relational database service based on SQL Server.

Key Features:

- Automatic backups and point-in-time restore
- Built-in high availability (99.99% SLA)
- Intelligent performance optimization
- Elastic pools for managing multiple databases
- Serverless compute tier available
- Hyperscale for large databases

When to use:

- Relational data with ACID requirements
- Migrating from SQL Server
- Need managed database without infrastructure overhead
- Applications requiring SQL compatibility

When NOT to use:

- NoSQL or document-based data (use Cosmos DB)
- Need specific SQL Server features not supported
- Simple key-value storage (use Table Storage)

Official Documentation: <https://learn.microsoft.com/en-us/azure/azure-sql/database/>



8. Azure Functions

What it is: Serverless compute service for event-driven code execution.

Key Features:

- Pay only for execution time
- Auto-scaling based on demand
- Multiple trigger types (HTTP, timer, queue, blob, Event Grid, etc.)
- Support for C#, JavaScript, Python, Java, PowerShell
- Durable Functions for stateful workflows
- Premium plan for VNET integration

When to use:

- Event-driven processing
- Scheduled tasks and automation
- Microservices and APIs with variable traffic
- Processing files, messages, or data streams
- Webhooks

When NOT to use:

- Long-running processes (use Durable Functions or containers)
- Complex stateful applications without Durable Functions
- Need predictable consistent performance

Official Documentation: <https://learn.microsoft.com/en-us/azure/azure-functions/>



9. Azure Virtual Network (VNet)

What it is: Private network infrastructure in Azure for resource connectivity.

Key Features:

- Isolation and segmentation with subnets
- Network Security Groups (NSGs) for traffic filtering
- VPN Gateway and ExpressRoute for hybrid connectivity
- VNet peering for connecting networks
- Service endpoints and private endpoints
- Azure Firewall and DDoS protection

When to use:

- Always - fundamental for most deployments
- Isolating resources and controlling traffic
- Connecting cloud and on-premises resources
- Implementing network security

When NOT to use:

- You really can't avoid it for production scenarios

Official Documentation: <https://learn.microsoft.com/en-us/azure/virtual-network/>

10. Azure Cosmos DB

What it is: Globally distributed, multi-model NoSQL database service.

Key Features:

- Turnkey global distribution
- Five consistency levels
- Multiple APIs (SQL/Core, MongoDB, Cassandra, Gremlin, Table)
- Guaranteed single-digit millisecond latency (P99)
- Automatic and instant scalability
- Serverless and autoscale options

When to use:

- Global applications needing low latency
- Document, key-value, graph, or column-family data
- IoT and real-time applications



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- Highly scalable NoSQL needs
- Gaming leaderboards and session storage

When NOT to use:

- Relational data best suited for SQL
- Budget-constrained projects (can be expensive)
- Simple storage needs (use Table Storage)

Official Documentation: <https://learn.microsoft.com/en-us/azure/cosmos-db/>

11. Azure Kubernetes Service (AKS)

What it is: Managed Kubernetes container orchestration service.

Key Features:

- Automated Kubernetes cluster management
- Integrated CI/CD and monitoring
- Built-in security and governance
- Auto-scaling (cluster and pods)
- Integration with Azure services
- Azure CNI and Kubenet networking

When to use:

- Microservices architectures
- Container-based applications at scale
- Need portability across clouds
- Complex orchestration requirements
- Service mesh implementations

When NOT to use:

- Simple applications (too complex)
- Team lacks Kubernetes expertise
- Single containerized app (use App Service or Container Instances)

Official Documentation: <https://learn.microsoft.com/en-us/azure/aks/>



12. Azure Key Vault

What it is: Secure storage for secrets, encryption keys, and certificates.

Key Features:

- Centralized secrets management
- Hardware Security Module (HSM) support
- Access control with managed identities
- Auditing and logging
- Certificate lifecycle management
- Integration with Azure services and apps

When to use:

- Always for production - never hardcode secrets
- Storing connection strings, API keys, passwords
- Managing encryption keys and certificates
- Compliance requirements (FIPS 140-2, PCI DSS)

When NOT to use:

- Development/testing with non-sensitive data (even then, it's good practice)

Official Documentation: <https://learn.microsoft.com/en-us/azure/key-vault/>

13. Azure Monitor & Application Insights

What it is: Comprehensive monitoring and diagnostics platform.

Key Features:

- Metrics and log collection
- Application performance monitoring (APM)
- Custom dashboards and alerts
- Log Analytics with KQL (Kusto Query Language)
- Integration across all Azure services
- Distributed tracing
- Live metrics stream

When to use:

- Always - monitoring is essential
- Performance troubleshooting



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- Understanding application behavior
- Creating operational dashboards and alerts
- Capacity planning

When NOT to use:

- You really should use it for any production workload

Official Documentation:

- Azure Monitor: <https://learn.microsoft.com/en-us/azure/azure-monitor/>
 - Application Insights:
<https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview>
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14. Azure DevOps / Azure Pipelines

What it is: Complete DevOps solution for planning, developing, delivering, and operating software.

Key Features:

- Azure Boards: Agile project management
- Azure Repos: Git repositories
- Azure Pipelines: CI/CD automation
- Azure Test Plans: Manual and exploratory testing
- Azure Artifacts: Package management
- Integration with GitHub and third-party tools

When to use:

- CI/CD pipeline implementation
- Source control and collaboration
- Agile project management
- Automated testing and deployment
- Multi-stage deployments

When NOT to use:

- Prefer GitHub exclusively (use GitHub Actions)
- Very simple deployment needs

Official Documentation: <https://learn.microsoft.com/en-us/azure/devops/>



15. Azure Logic Apps

What it is: Low-code/no-code workflow automation and integration service.

Key Features:

- Visual designer for workflows
- 400+ pre-built connectors
- Integration with SaaS and enterprise applications
- Trigger-based execution
- Built-in error handling and retry logic
- Standard and Consumption pricing models

When to use:

- Business process automation
- System integration without code
- Scheduled workflows
- API orchestration
- Event-driven automation
- ETL processes

When NOT to use:

- Complex custom logic (use Functions)
- High-throughput real-time processing
- Need full programmatic control

Official Documentation: <https://learn.microsoft.com/en-us/azure/logic-apps/>

Learning Path Recommendation

Beginner (Weeks 1-4):

1. Microsoft Entra ID - Identity foundation
2. Storage Account - Simplest to understand
3. Virtual Machines - IaaS fundamentals
4. Virtual Network - Connectivity basics

Intermediate (Weeks 5-8):

5. App Service - PaaS concepts
6. SQL Database - Managed databases
7. Key Vault - Security best practices



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8. Azure Monitor - Observability

Advanced (Weeks 9-12):

- 9. Functions - Serverless architecture
- 10. Cosmos DB - NoSQL and global distribution
- 11. Azure OpenAI - AI integration
- 12. Azure AI Services - Pre-built AI capabilities

Expert (Weeks 13-16):

- 13. AKS - Container orchestration
- 14. Azure DevOps - CI/CD practices
- 15. Logic Apps - Workflow automation

These 15 services cover the core pillars of Azure: compute, storage, networking, databases, security, identity, AI, monitoring, and DevOps. Master these, and you'll be well-equipped to architect and implement comprehensive Azure solutions.